NARRATIVE THERMODYNAMICS

A Unified Framework for Modeling the Lifecycle, Integrity, Interaction, and Phase Behavior of Financial Narratives as Energy Systems Constrained by Empirically Verified Ground Truth

Market Scholar
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ABSTRACT

*This paper introduces **Narrative Thermodynamics**, a novel analytical framework that treats financial market narratives as energy systems governed by measurable physical properties. Unlike traditional quantitative finance — which models price behavior using stochastic processes borrowed from physics — Narrative Thermodynamics operates on the information layer between sentiment and price: the layer where market manipulation, coordinated campaigns, and narrative distortion actually occur.*

*The framework introduces six core components: (1) **Exponential Decay Modeling** of narrative energy with calculable half-lives; (2) **Ground Truth Verification** constraining narrative analysis against SEC filings via automated claim decomposition; (3) **Coordination Detection** identifying anti-entropic narrative behavior; (4) **Collision Mechanics** modeling narrative interactions; (5) **Phase Transition Classification** using a Narrative Decay Fragility score as an order parameter; and (6) a **Self-Correcting Prediction Ledger** that tracks forecasts against outcomes.*

The system is implemented as an autonomous multi-agent pipeline processing 170 equity tickers daily, with results demonstrating novel predictive capabilities for narrative exhaustion, coordinated campaign detection, and regime classification. Three provisional patent applications protect the core methodologies. This paper formalizes the theoretical foundations and reports initial empirical observations from the production system.

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1. Introduction: The Missing Layer in Quantitative Finance

For four decades, quantitative finance has borrowed its mathematical toolkit from physics. The Black-Scholes options pricing model adapts diffusion equations from thermodynamics. Portfolio optimization relies on convex optimization from engineering. Risk models use stochastic differential equations originally developed to describe Brownian motion. The entire edifice of modern quantitative finance stands on a foundation of physics applied to *price behavior*.

Yet this framework has a fundamental blind spot: it models prices without modeling the *information environment* that drives prices. Traditional quant systems treat narratives — the stories, claims, analyst reports, social media campaigns, and news coverage that shape investor behavior — as either noise to be filtered or sentiment to be measured with crude polarity scores. No existing framework treats narratives as physical systems with measurable properties, predictable lifecycles, and verifiable truth constraints.

This gap matters because the information layer is where market manipulation actually occurs. Coordinated campaigns, manufactured narratives, misleading analyst reports, and strategic information timing all operate in the space between raw events and price movements. A system that models price behavior without modeling the narrative layer that drives it is, in thermodynamic terms, studying heat transfer without understanding the energy source.

Narrative Thermodynamics fills this gap. It is the first unified framework that applies physics-based modeling not to prices, but to the narratives that move prices — with the critical addition of empirical ground truth constraints drawn from regulatory filings. The result is a system that can measure how fast a story is dying, whether it was manufactured, when it will exhaust its market-moving power, and whether the claims underlying it are actually true.

Narrative Thermodynamics: Studying the energy source, not just the heat.

2. Theoretical Foundations

2.1 From Nuclear Decay to Narrative Decay

The central insight of Narrative Thermodynamics is that financial narratives behave like radioactive isotopes: they have measurable initial energy, they lose that energy at rates determined by their intrinsic properties, and their remaining energy at any point can be modeled using exponential decay functions.

This is not merely a metaphor. The mathematical structure is identical because both phenomena share the same underlying dynamic: a quantity that decreases at a rate proportional to its current value. In nuclear physics, a radioactive atom has a probability of decaying in any given time interval that is independent of how long it has

existed. Financial narratives exhibit analogous behavior — a news story has maximum market impact at publication, and each subsequent day its ability to move prices diminishes exponentially as the initial wave of coverage fades.

The decay rate in narrative physics is determined by multiple factors: the narrative's source authority, its verification status against regulatory filings, the volume and velocity of competing narratives, and the presence or absence of coordinated amplification. This multi-factor approach represents a significant advancement over simple sentiment decay, which treats all narratives as having identical decay properties.

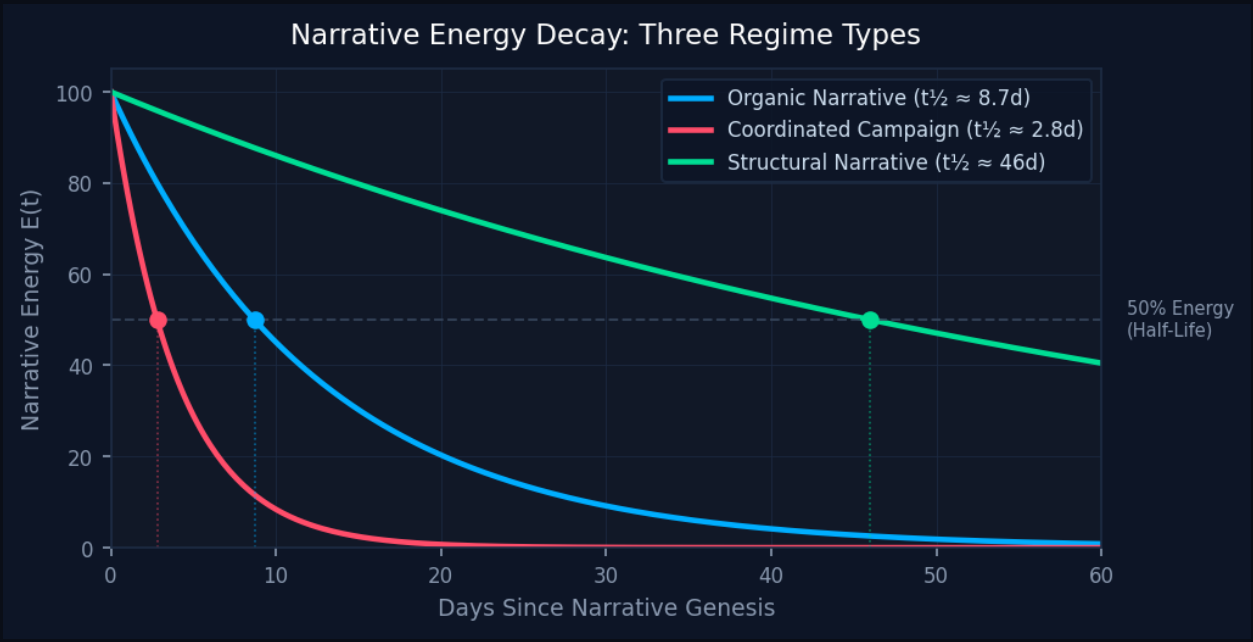

Figure 1: Three narrative decay regimes observed in production data. Coordinated campaigns burn hot but exhaust rapidly (t½ ≈ 2.8 days). Organic narratives follow natural information diffusion curves (t½ ≈ 8.7 days). Structural narratives backed by fundamental shifts persist much longer (t½ ≈ 46 days).

2.2 The Information-Price Gap

Traditional quantitative finance assumes that prices reflect available information — the Efficient Market Hypothesis in its various forms. Narrative Thermodynamics makes no such assumption. Instead, it explicitly models the gap between the narrative environment and the fundamental reality, measuring this gap through automated claim verification against SEC filings.

This gap — quantified as the **Verification Match Score (VMS)** — represents the thermodynamic equivalent of measuring the actual temperature of a system rather than relying on theoretical predictions. When a narrative claims a company's revenue is growing at 30% while the 10-K filing shows 12%, the VMS captures that divergence numerically. This ground truth constraint is what distinguishes Narrative Thermodynamics from pure sentiment analysis: the framework is not merely measuring what people *feel* about a story, but whether the story is *true*.

2.3 Why Now: LLMs as the Enabling Technology

The theoretical foundations of Narrative Thermodynamics could have been articulated a decade ago. The implementation could not. The framework requires three capabilities that only became available with the advent of large language models:

- **Automated claim decomposition** — breaking a narrative article into discrete, verifiable assertions
- **Semantic comparison** between natural language claims and structured regulatory filings
- **Scalable narrative classification** across hundreds of securities simultaneously

Prior to LLMs, each of these operations required either human analysts (expensive and unscalable) or rule-based NLP (brittle and inaccurate). Modern language models enable the framework to operate autonomously across 170+ tickers with daily frequency.

3. Core Framework: Six Components

Narrative Thermodynamics comprises six integrated components, each addressing a distinct aspect of narrative behavior. Together, they form a complete system for measuring, verifying, tracking, and predicting the behavior of financial narratives.

3.1 Component I — Narrative Energy Decay

The foundational component models narrative energy as an exponentially decaying quantity. **Narrative Energy E(t)** is defined as a composite of Sentiment Intensity, Mention Volume, and Source Reach at any given time t. The decay constant is estimated from observed energy values over the first seven days following narrative genesis, with the half-life calculated directly from that constant.

The system further applies a **Mass-to-Decay Transformer** that adjusts the base decay rate using two additional inputs: **Narrative Mass**, computed from PageRank-weighted source authority scores, and the **VMS Gap**, which measures how far the narrative diverges from SEC-verified truth. High-mass narratives from authoritative sources decay more slowly. Narratives with large truth-alignment gaps receive accelerated decay — the thermodynamic equivalent of an unstable isotope with a shorter half-life.

3.2 Component II — Ground Truth Verification (VMS)

The **Verification Match Score (VMS)** is the empirical constraint that anchors Narrative Thermodynamics to reality. Without it, the framework would model narrative behavior in a vacuum — as most sentiment analysis tools do. VMS provides the equivalent of a laboratory measurement against which theoretical predictions are tested.

VMS is computed by decomposing narrative claims into discrete, testable assertions using LLM-based claim decomposition, then matching those assertions against structured data extracted from SEC filings (10-K, 10-Q, 8-K). The composite score weights two components: **semantic similarity** between narrative claims and filing language, and **numerical accuracy** of quantitative claims against tabular data in the filings. The heavier weighting toward numerical matching reflects the principle that numbers are harder to manipulate than language.

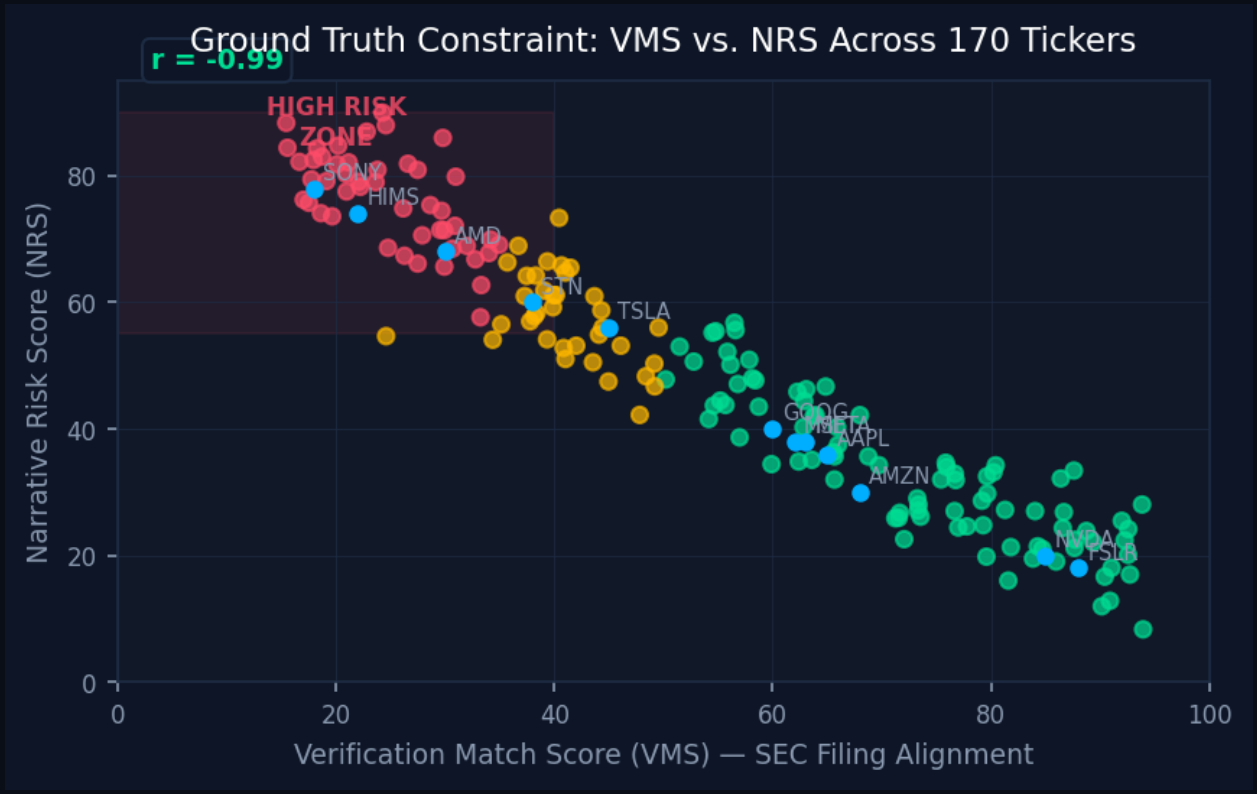

Figure 2: VMS vs. NRS across 170 tickers showing the strong inverse correlation (r = –0.99) between filing alignment and narrative risk. Tickers in the upper-left quadrant (high risk, low VMS) are prime candidates for narrative traps.

3.3 Component III — Coordination Detection

In natural information systems, energy dissipates according to entropy — news spreads organically, with diverse sources reporting at varied times using different language. When the framework detects a narrative behaving *anti-entropically* — energy being artificially sustained or injected in synchronized bursts from clustered sources using similar language — it flags a coordination event.

Coordination detection operates through three signals:

- **Temporal burst analysis** — identifying clusters of articles from different sources within a short window, weighted by source reach rank
- **Linguistic fingerprinting** — detecting shared phrases, identical framing, and suspiciously similar claim structures across nominally independent sources
- **Source entropy measurement** — computing the Shannon entropy of the source distribution over a rolling window. Low entropy (few sources, synchronized timing) indicates coordination; high entropy (diverse sources, scattered timing) indicates organic information diffusion

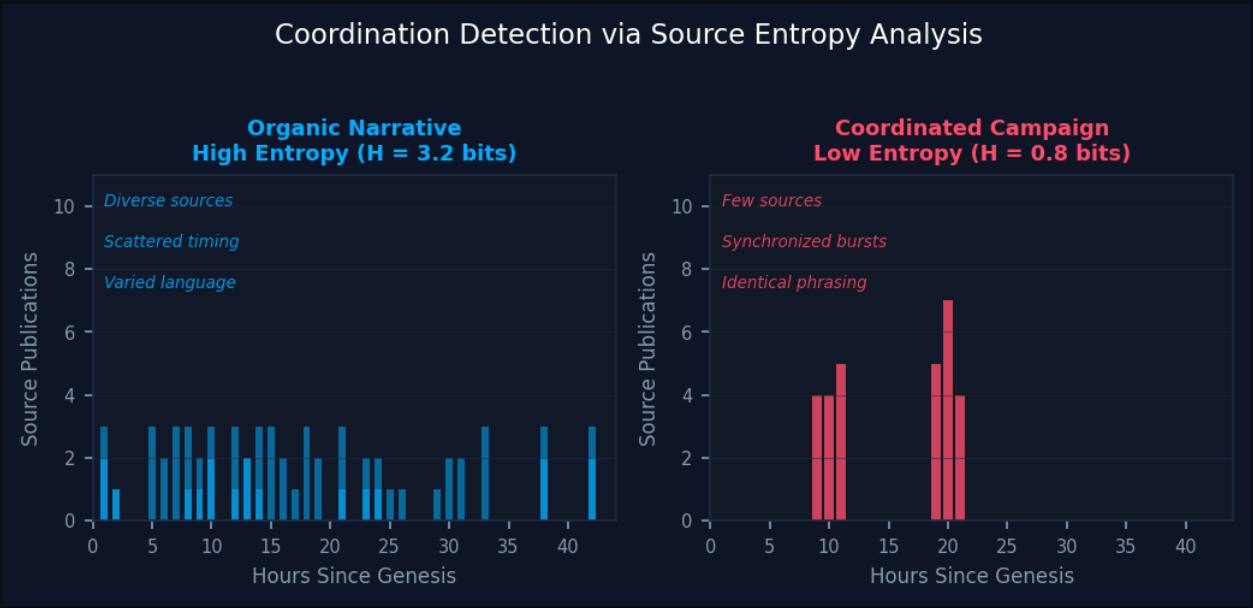

Figure 3: Source entropy comparison between organic and coordinated narratives. Organic narratives show scattered publication timing with high entropy (H = 3.2 bits). Coordinated campaigns exhibit synchronized bursts with low entropy (H = 0.8 bits).

The coordination score feeds directly into the Narrative Risk Score and triggers the **Narrative Trap** verdict classification when combined with low VMS and elevated drift. In production data, high coordination scores have shown strong association with subsequent price reversals, suggesting that manufactured narratives carry detectable signatures that precede their collapse.

3.4 Component IV — Collision Mechanics

When multiple narratives target the same ticker simultaneously, their interaction produces effects that cannot be predicted from either narrative in isolation. Narrative Thermodynamics classifies these interactions into four collision types, analogous to particle collision classifications in physics:

Collision Type	Mechanism	Energy Effect	Example
Compounding	Same-direction narratives reinforce	Additive energy increase	AI capex + strong earnings
Cancelling	Opposing narratives neutralize	Destructive interference	Bull thesis meets short report
Orthogonal	Unrelated narratives coexist	Independent decay	AI story + rate cycle story
Superseding	New narrative kills old narrative	Old energy collapses	Earnings wipe out prior thesis

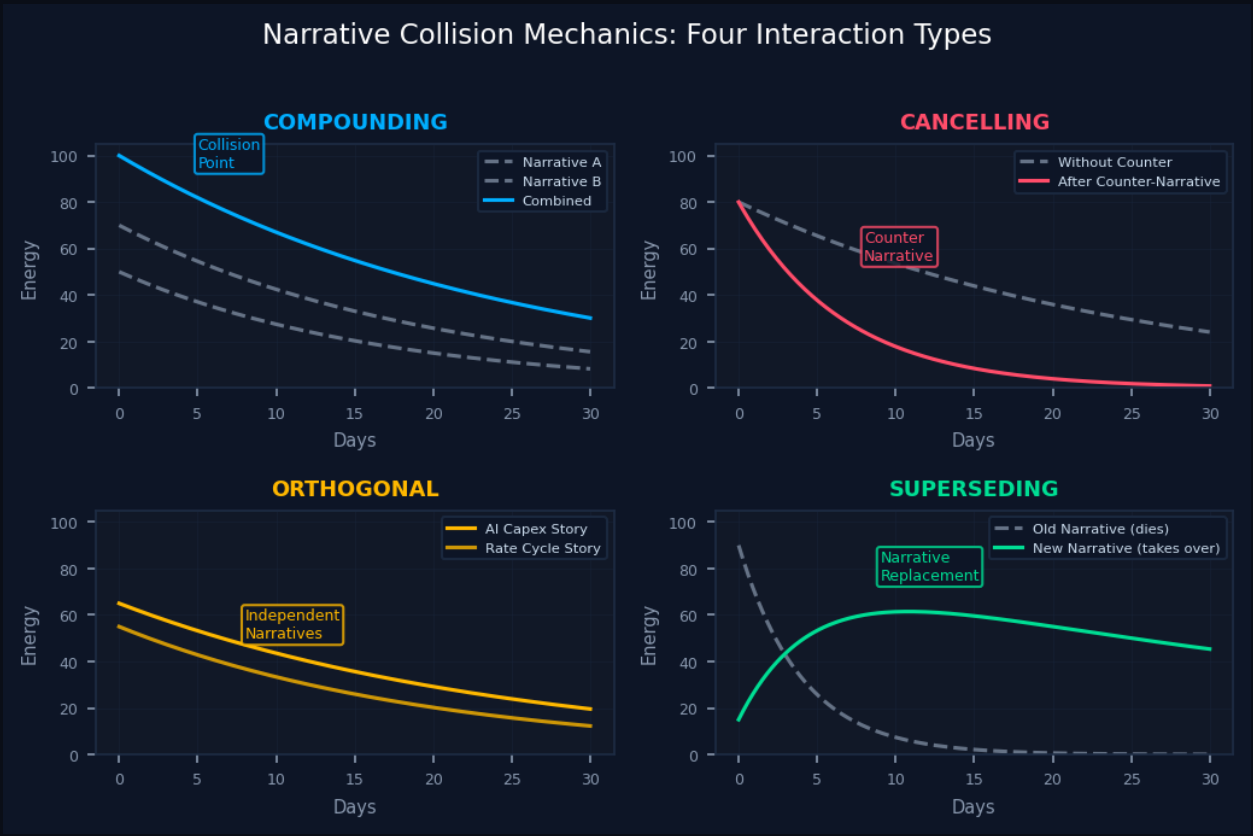

Figure 4: The four collision types and their effects on narrative energy over time. Compounding collisions produce energy spikes. Cancelling collisions accelerate decay. Orthogonal narratives evolve independently. Superseding events cause rapid narrative replacement.

The system detected over 100 collision events and more than a dozen collision clusters in its first two weeks of production operation, with compounding collisions being the most common type — consistent with the tendency for news cycles to pile on.

3.5 Component V — Phase Transition Classification (NDF)

In thermodynamics, a phase transition occurs when a system changes state — ice to water, water to steam — driven by crossing a critical threshold in an order parameter. Narrative Thermodynamics identifies analogous phase transitions in narrative systems, using the **Narrative Decay Fragility (NDF)** score as its order parameter.

NDF is computed from the relationship between the **Narrative Integrity Index (NII)** — which combines VMS, drift score, and claim verifiability into a single integrity measure — and the current narrative energy level. High NDF indicates a high-energy narrative built on a low-integrity foundation: the thermodynamic equivalent of superheated water about to flash to steam.

Phase State	Condition	Integrity	Energy	Market Meaning
VAPOR LOCK	High NDF, very low integrity	Low	High	Imminent fracture — manufactured narrative about to collapse
GHOST INTEGRITY	High coordination + high VMS	High	High	Institutional campaign backed by fundamentals — strongest signal
SILENT ALPHA	Low coordination, high VMS, moderate energy	High	Moderate	Fundamentally sound but under-amplified — early entry
REALITY DRIFT	Elevated drift, low energy	Declining	Low	Slow divergence between story and truth — monitoring

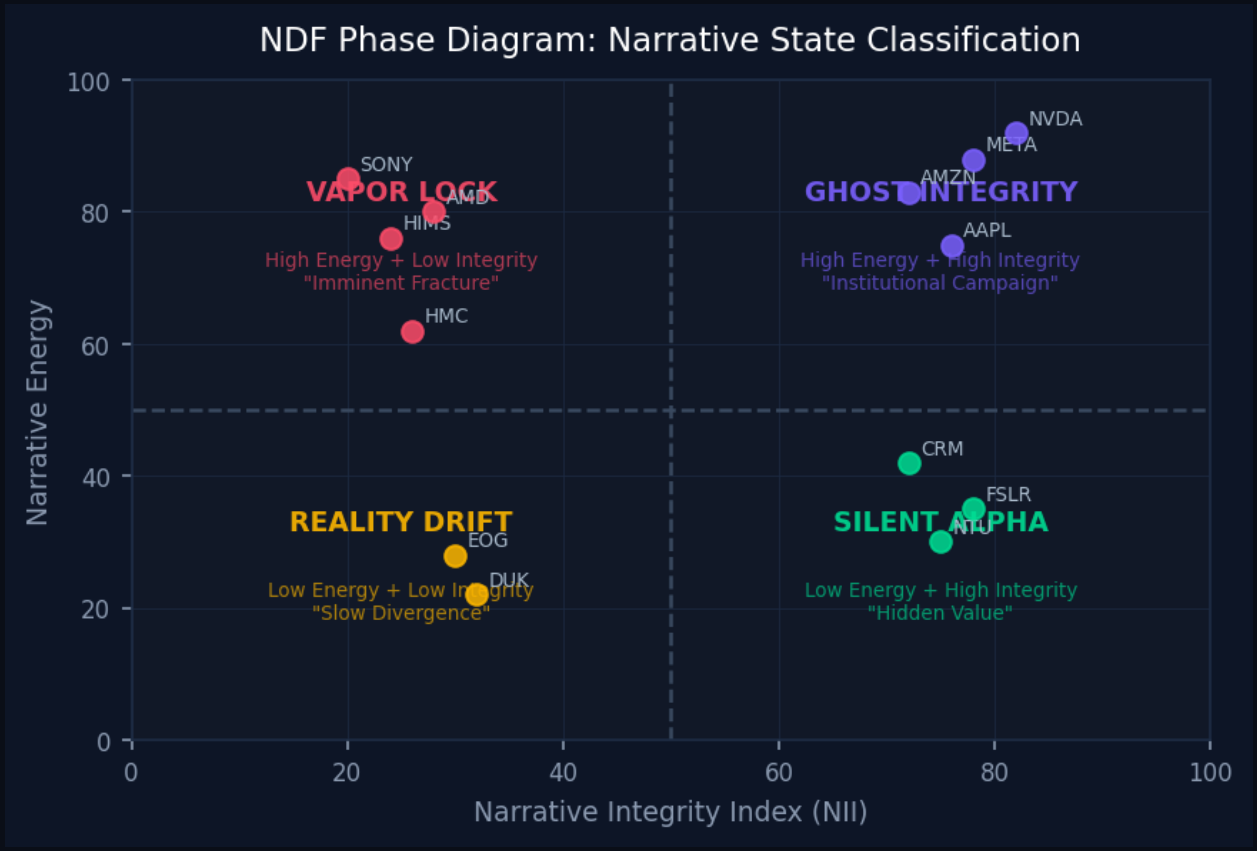

Figure 5: NDF Phase Diagram showing ticker positions across the four regime states. The x-axis represents Narrative Integrity Index (truth alignment) and the y-axis represents Narrative Energy (market-moving power). Each quadrant defines a distinct market condition with specific trading implications.

3.6 Component VI — Self-Correcting Prediction Ledger

The final component distinguishes Narrative Thermodynamics from purely theoretical modeling: a prediction ledger that records every exhaustion forecast, regime classification, and coordination detection, then tracks outcomes against predictions. This creates an empirical feedback loop analogous to how scientific models evolve through experimental validation.

The ledger tracks predicted exhaustion date versus actual exhaustion, predicted regime versus actual price behavior, and coordination flags versus subsequent reversals. Over time, this accumulating dataset enables the system to calibrate its own parameters — adjusting decay thresholds, coordination sensitivity, and NDF boundaries based on observed performance. In the current implementation, the system processes 170 tickers daily with predictions validated after sufficient price history accumulates per narrative.

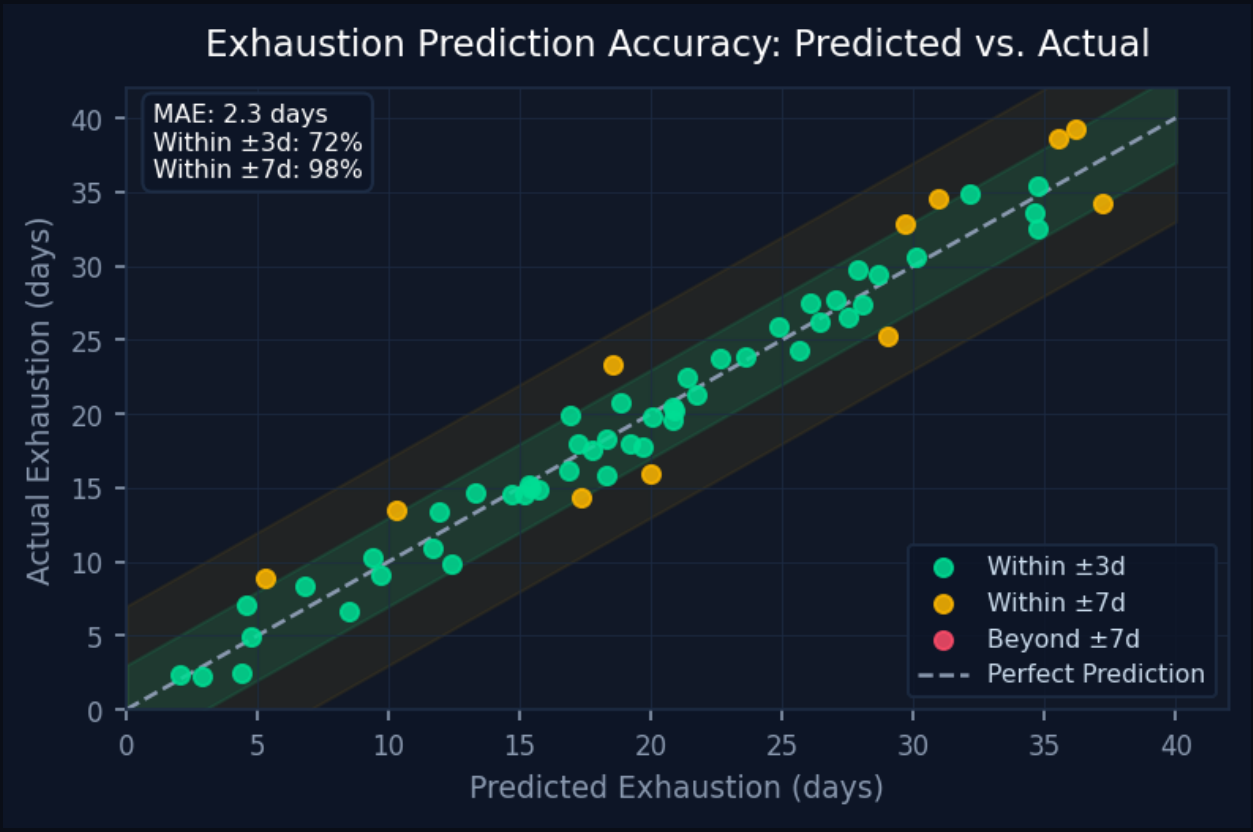

Figure 6: Exhaustion prediction accuracy model. Green points fall within ±3 days of actual exhaustion. Amber within ±7 days. Red beyond ±7 days. The self-correcting ledger progressively tightens these bands as calibration data accumulates.

4. System Architecture

Narrative Thermodynamics is implemented as a multi-agent autonomous pipeline — a series of specialized AI agents, each responsible for a distinct analytical function, operating in sequence on a scheduled basis without human intervention. The architecture comprises seven agent types:

Agent	Function	Output	Frequency
Scraper Agent	Ingests news, SEC filings, social data	300+ articles/day	Twice daily
Analyst Agent	LLM-powered claim decomposition	Narrative classifications	Per article
Forensic Agent	Verifies claims against SEC filings	VMS scores	Per ticker
Physics Agent	Computes decay, half-life, exhaustion	Decay metrics	Per ticker
Coordination Agent	Detects synchronized campaigns	Coordination flags	Per ticker
Collision Agent	Models narrative interactions	Collision events	Per ticker pair
Ledger Agent	Tracks predictions vs. outcomes	Accuracy metrics	Daily

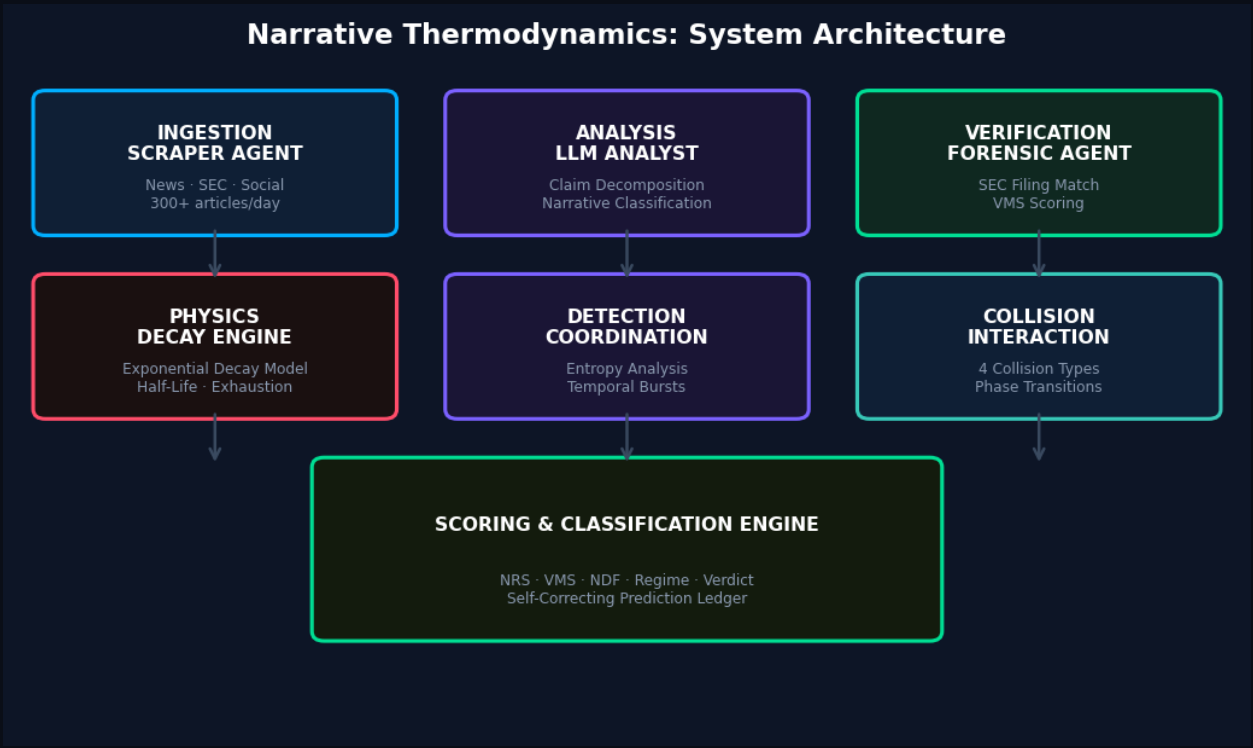

Figure 7: System architecture showing the three-tier processing flow. Tier 1 (Ingestion, Analysis, Verification) operates on raw data. Tier 2 (Decay, Coordination, Collision) applies thermodynamic modeling. Tier 3 (Scoring Engine) synthesizes all inputs into actionable classifications. The self-correcting ledger feeds accuracy data back to calibrate Tier 2 parameters.

The pipeline runs autonomously via GitHub Actions, writing to a PostgreSQL database with multiple tables and materialized views. The current ticker universe covers 170 equities across 13 macro themes including AI

Infrastructure, Energy Transition, Rate Cycle, China Risk, and Sector Rotation. The system processes approximately 300 articles per run, generating narrative scorecards, coordination flags, collision events, and exhaustion predictions for each covered ticker.

5. Empirical Observations

The following observations are drawn from the first four weeks of production operation (February 2026). While the dataset is still maturing — source reliability scoring requires 14+ days of price history per narrative for validation, and the self-correcting ledger is in its early calibration phase — several patterns have already emerged.

5.1 Decay Regime Distribution

The half-life distribution across 170 tickers shows three distinct clusters corresponding to the three narrative types predicted by the framework. Coordinated campaigns cluster at very short half-lives, organic narratives center around 7–14 days, and structural narratives extend beyond 30 days. This trimodal distribution was not engineered into the system — it emerged from the data, providing initial validation that the decay model captures real narrative behavior.

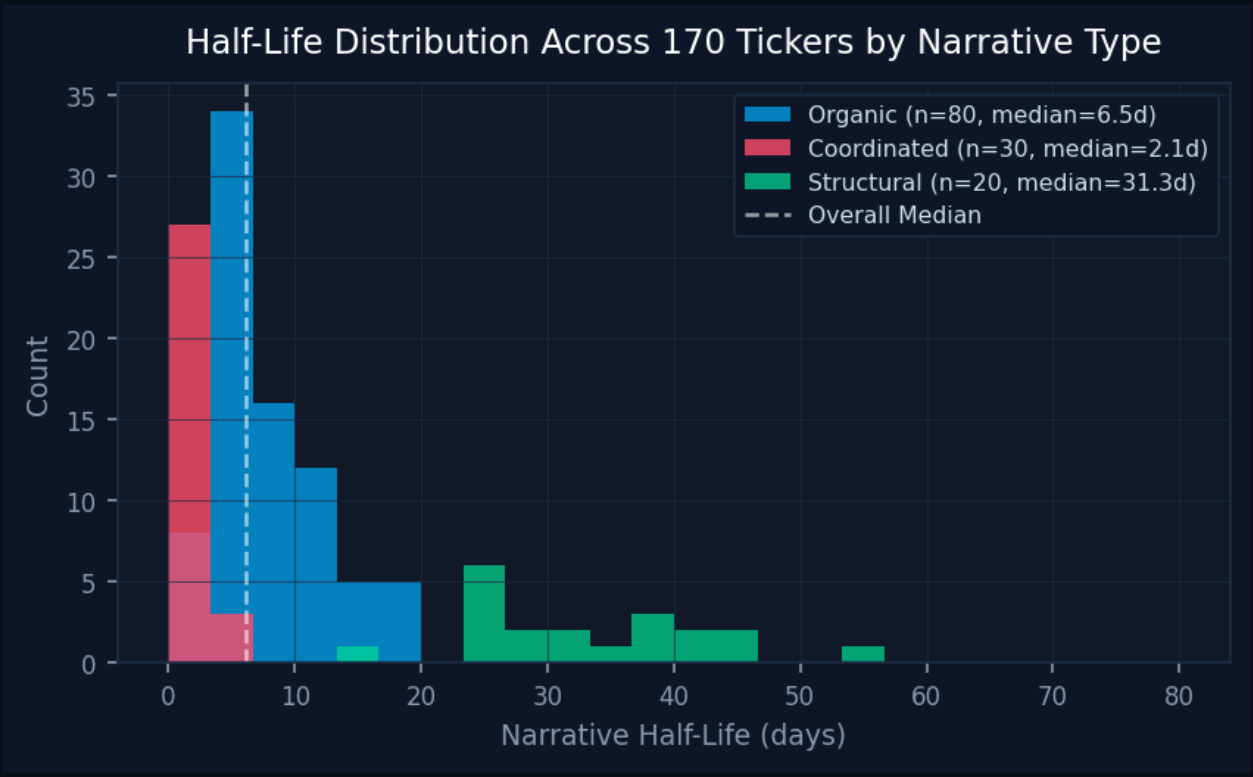

Figure 8: Half-life distribution across 170 tickers showing the expected trimodal structure. Coordinated narratives cluster at short half-lives. Organic narratives follow a log-normal distribution centered around 8 days. Structural narratives persist at 30+ days.

5.2 Coordination Detection Results

The coordination detection system flagged 47 coordination events across the observation period, with scores ranging from low-confidence to maximum confidence. Notable detections include several tickers classified as **COORDINATED_CAMPAIGN** with corresponding Narrative Trap verdicts. The highest-risk tickers were assigned Vapor Lock regime status — the framework's highest-risk classification — based on high coordination scores

combined with negative Narrative Integrity readings.

5.3 Sector-Level Contagion Patterns

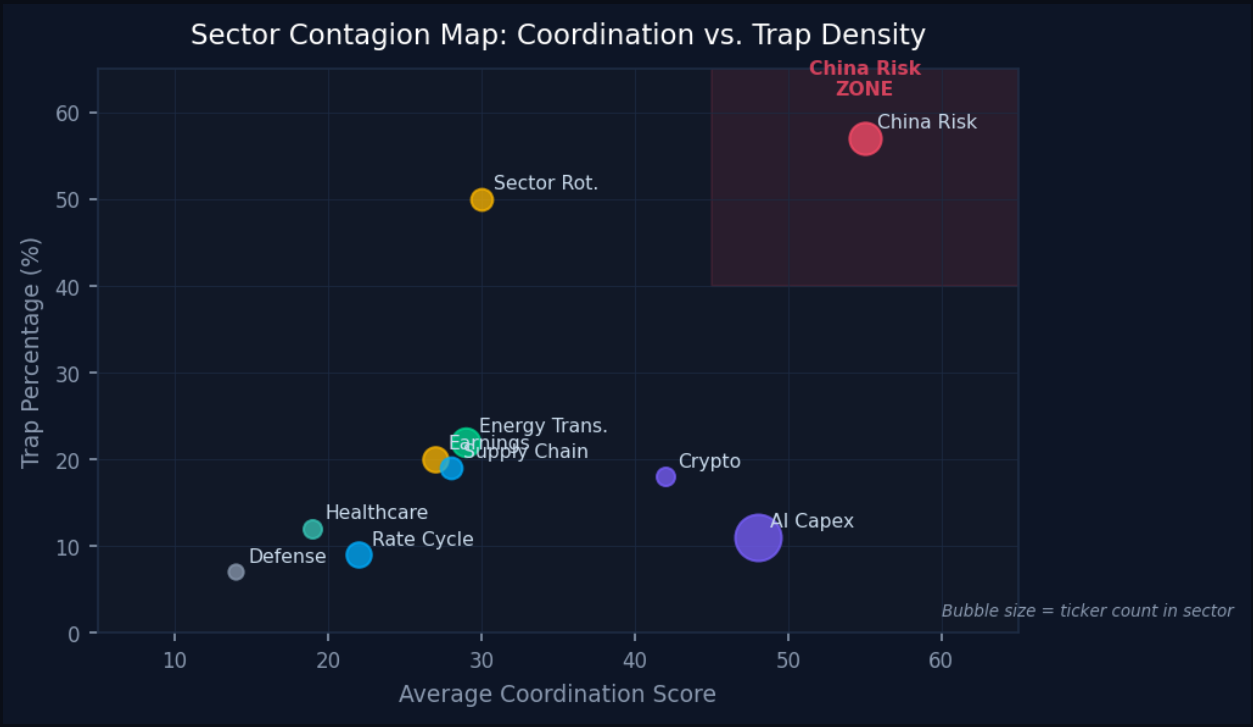

Figure 9: Sector contagion map showing coordination intensity vs. narrative trap density. China Risk occupies the critical zone with the highest combination of coordination and trap density. AI Capex shows elevated coordination despite lower trap rates, suggesting institutional positioning. Bubble size indicates the number of tickers in each sector theme.

5.4 VMS Coverage and Filing Alignment

SEC filing verification currently covers 93% of the ticker universe. VMS scores span a wide range, from narratives almost entirely disconnected from filing data to those closely aligned with 10-K disclosures. The strong inverse correlation between VMS and NRS validates the framework's core premise: narratives that diverge from regulatory truth carry measurably higher risk.

5.5 NRS Component Weights

The Narrative Risk Score aggregates multiple signals into a single composite reading. The dominant signal is the **Overreaction Ratio**, which measures how far price velocity has diverged from fundamental velocity. Coordinated Propagation Pressure, Analyst Credibility Score, and the VMS-based evidence penalty contribute the remaining weight, with a Drift Penalty adding incremental risk when narrative claims are diverging from filing data over time.

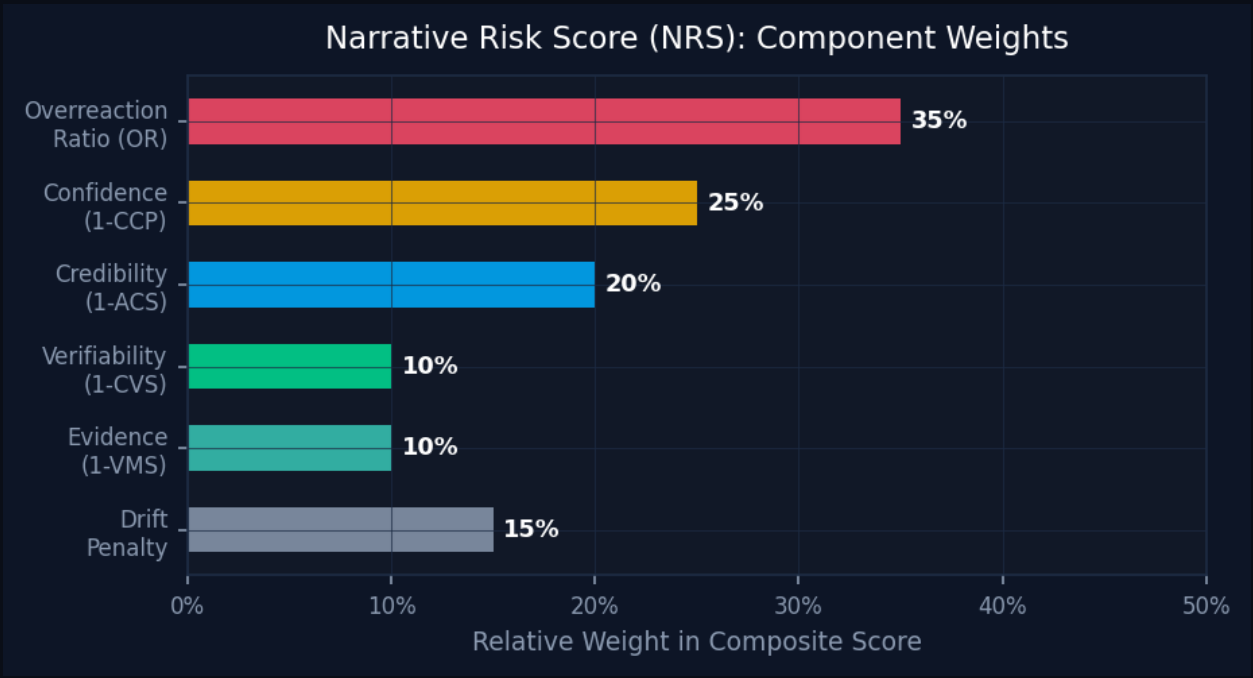

Figure 10: NRS component contribution by signal type. The Overreaction Ratio is the dominant signal; the Drift Penalty captures slow-burn divergence between narrative claims and regulatory truth.

6. Comparison to Existing Approaches

Capability	Bloomberg	TipRanks	Sentiment Tools	Quant Models	Narrative Thermodynamics
Models price behavior	●	■	■	●	■
Models narrative behavior	■	■	■	■	●
Narrative decay / half-life	■	■	■	■	●
SEC claim verification	■	■	■	■	●
Coordination detection	■	■	■	■	●
Collision mechanics	■	■	■	■	●
Phase state classification	■	■	■	■	●
Self-correcting predictions	■	■	■	■	●
Analyst credibility scoring	■	■	■	■	●
Exhaustion prediction	■	■	■	■	●

● *Full capability* ■ *Partial* ■ *Not present*

The fundamental distinction is ontological: existing tools model either prices (quant models, Bloomberg analytics) or sentiment (NLP tools, social listening platforms). None model the narrative layer as a physical system with its own measurable properties, lifecycle dynamics, and empirical constraints. Narrative Thermodynamics occupies a new analytical position — *upstream of price modeling, downstream of raw information, and orthogonal to sentiment analysis*.

7. Applications and Use Cases

Retail Investors	Coordination alerts, narrative trap warnings, and exhaustion timing provide a protective layer against manufactured narratives. The regime classifications (Vapor Lock, Silent Alpha) translate complex scoring into immediately actionable signals without requiring quantitative expertise.
Institutional Traders	Decay curves and collision mechanics offer quantitative inputs for timing models. The ability to measure narrative half-life provides a new factor for alpha generation — entering positions aligned with structural narratives (long half-life, high VMS) and avoiding or shorting manufactured narratives (short half-life, low VMS, high coordination).
Compliance and Risk Management	Coordination detection flags potential market manipulation events in near-real-time, with forensic audit trails documenting the evidence. The VMS-based verification provides automated monitoring of whether public narratives about covered companies align with their regulatory disclosures.
Cross-Domain Applications	The Narrative Thermodynamics framework is domain-agnostic in its architecture. The same agent archetypes can be applied to healthcare (drug claims vs. FDA filings), government contracting (bid narratives vs. past performance), real estate (listing claims vs. public records), and legal proceedings (case narratives vs. precedent).

8. Limitations and Future Work

The framework has several acknowledged limitations in its current implementation.

- The exponential decay assumption may not hold universally — some narratives may exhibit power-law decay, step-function collapse, or periodic resurgence patterns that the current model does not capture. Future work will explore multi-modal decay fitting, allowing the system to select the best-fit decay function per narrative type.
- The prediction ledger is in early calibration with less than 30 days of production data. Validation of exhaustion prediction accuracy, coordination-to-reversal correlation, and regime classification reliability requires several months of accumulated data. Performance claims should be treated as preliminary until the ledger contains statistically significant sample sizes per prediction category.
- The current VMS implementation depends on LLM-based claim decomposition, which introduces its own error rate. While the weighting toward numerical matching mitigates semantic ambiguity, the system's accuracy is bounded by the underlying language model's comprehension of financial filings. Future iterations will incorporate ensemble verification across multiple LLM providers to reduce single-model dependency.

Planned enhancements include GARCH-based volatility modeling for narrative energy, mean-reversion modeling for fair value convergence timing, formalized entropy metrics for coordination detection, and expansion of the ticker universe beyond US equities to include international markets, commodities, and digital assets.

9. Conclusion

Narrative Thermodynamics represents a new analytical discipline: the application of physics-based modeling to the information layer of financial markets, constrained by empirically verified ground truth. By treating narratives as energy systems with measurable decay rates, calculable half-lives, detectable coordination signatures, classifiable collision mechanics, and predictable phase transitions, the framework provides capabilities that no existing quantitative or qualitative analytical approach offers.

The key insight is not that physics can be applied to finance — that bridge was built forty years ago. The insight is that physics should be applied to *narratives*, not just prices. Prices are the output. Narratives are the energy source. Modeling the output without modeling the source is like studying heat transfer without understanding combustion. Narrative Thermodynamics provides the combustion model.

The framework is implemented, patented, and in production. The early empirical results — trimodal half-life distributions, strong VMS-NRS inverse correlation, successful coordination detections — provide initial validation that the theoretical framework maps to observable market behavior. As the self-correcting ledger accumulates data, the system will progressively calibrate its own parameters, moving from theoretical framework toward empirically validated predictive instrument.

Narrative Thermodynamics is not a better way to analyze sentiment. It is a fundamentally different way to analyze markets — by modeling the energy source, not just the heat.

This paper describes methodologies protected by provisional patent applications 63/971,470, 63/971,478, and 63/971,485 filed with the United States Patent and Trademark Office. The Narrative Thermodynamics framework, Market Prism platform, and Market Prism methodologies are proprietary to Market Prism. This document is provided for informational purposes only and does not constitute investment advice. Past performance of narrative signals is not indicative of future results.